

# REAL WORLD EVIDENCE STUDY REPORT

Disease: Type 2 Diabetes (Proxy for MG Pipeline Demo)

|                |                                                        |
|----------------|--------------------------------------------------------|
| Study Type:    | Retrospective Cohort Study — Descriptive               |
| Data Source:   | Synthea Synthetic EMR v3.x · Massachusetts · n = 5,000 |
| Analysis Date: | 2026-05-13                                             |
| Pipeline:      | Bronze → Silver → Gold (PostgreSQL + Python)           |
| Status:        | ■ ■ Synthetic Data Only — Not for Clinical Use         |

## 1. Study Objectives

This study aimed to identify and characterise a patient cohort from synthetic EMR data using a standard pharmacoepidemiology pipeline. Target disease: Type 2 Diabetes (Proxy for MG Pipeline Demo). Objectives: (i) apply SNOMED CT → ICD-10-CM clinical code mappings to identify eligible patients; (ii) compute baseline demographics and comorbidity burden via the Charlson Comorbidity Index (CCI); (iii) characterise healthcare utilisation patterns.

## 2. Methods

### 2.1 Data Source

Synthea™ (MITRE Corporation) open-source synthetic patient simulator. Population of 5,000 patients representing Massachusetts, USA. Data domains: diagnoses, medications, procedures, observations, encounters. Data is clinically realistic but entirely de-identified and synthetic.

### 2.2 Phenotyping Algorithm

Patients were identified using SNOMED CT codes mapped to ICD-10-CM equivalents via the NLM reference mapping. A minimum of two coded encounters bearing the index disease code was required for inclusion — this two-encounter rule reduces misclassification bias from incidental or rule-out coding, consistent with published pharmacoepidemiology guidelines.

### 2.3 Comorbidity Assessment — Charlson Comorbidity Index

The CCI was calculated per patient from conditions recorded in the EMR. Seventeen comorbid conditions are assigned weights of 1, 2, 3, or 6 and summed. Higher CCI indicates greater comorbidity burden and predicts 10-year mortality risk. Widely used as a covariate in pharmacoepidemiology regression models.

### 2.4 Statistical Analysis

Descriptive statistics only. Continuous variables: median and mean. Categorical variables: count and percentage. No inferential testing performed.

## 3. Results

| Characteristic            | Value       |
|---------------------------|-------------|
| POPULATION                |             |
| Total patients in cohort  | 433         |
| Deceased at analysis date | 161 (37.2%) |
|                           |             |

| DEMOGRAPHICS                      |             |
|-----------------------------------|-------------|
| Median age at index (years)       | 46.6        |
| Mean age at index (years)         | 48.3        |
| Female                            | 202 (46.7%) |
| Male                              | 231 (53.3%) |
| COMORBIDITY BURDEN (CCI)          |             |
| Median Charlson Comorbidity Index | 3.0         |
| Mean Charlson Comorbidity Index   | 3.32        |
| CCI = 0 (Low risk)                | 0 (0.0%)    |
| CCI 1–2 (Mild)                    | 86 (19.9%)  |
| CCI 3–4 (Moderate)                | 274 (63.3%) |
| CCI ≥ 5 (Severe)                  | 73 (16.9%)  |
| HEALTHCARE UTILISATION            |             |
| Median total encounters           | 150         |
| Median hospitalisations           | 5           |
| Median unique medications         | 12          |

Table 1. Baseline characteristics of the study cohort.

## 4. Limitations

Analysis based entirely on Synthea synthetic data — does not reflect real patient populations. Synthea is calibrated for common diseases; rare diseases (e.g. Myasthenia Gravis) are under-represented, necessitating use of a proxy condition for pipeline demonstration. In a production study, methods would be applied to validated real-world sources such as CPRD, Optum Clinformatics, IBM MarketScan, or MIMIC-IV.

## 5. Conclusions

A complete end-to-end RWE cohort pipeline was successfully implemented, identifying 433 patients meeting the Type 2 Diabetes (Proxy for MG Pipeline Demo) phenotyping criteria (median age 47 yrs, median CCI 3, median 150 total encounters). The pipeline demonstrated full data flow from raw EMR ingestion through bronze/silver/gold layers, applying SNOMED CT → ICD-10-CM mappings, computing the Charlson Comorbidity Index, and producing an interactive Plotly Dash dashboard and this structured study report — replicating the foundational methodology of published pharmacoepidemiology studies.